

VELIZY-VILLACOUBLAY AB, France

WMO: 071470

Lat: 48.770N

Lon: 2.205E

Elev: 577

StdP: 14.39

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.6	26.5	15.2	12.1	27.6	19.3	14.8	29.6	25.5	48.2	23.0	47.0	7.2	30	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	86.7	67.7	83.0	66.2	79.7	64.9	69.7	82.7	67.9	79.3	66.3	76.7	7.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	95.9	74.3	63.8	90.6	72.5	62.3	85.7	70.8	34.0	82.8	32.5	79.2	31.2	76.9	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
20.4	18.0	16.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			19.7	93.4	3.9	4.3	16.9	96.5	14.6	99.0	12.4	101.4	9.6	104.6
			18.6	73.2	4.1	2.0	15.7	74.7	13.3	75.8	11.0	77.0	8.0	78.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual														
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	39.3	40.7	46.1	50.7	57.2	63.1	66.9	66.7	60.5	54.0	45.5	40.4	(6)
(7)		DBStd	11.71	6.99	6.63	6.29	6.40	6.45	6.44	6.30	6.11	5.55	6.09	6.12	6.72	(7)
(8)		HDD50	1316	333	262	153	66	11	1	0	0	1	30	158	301	(8)
(9)		HDD65	4802	796	679	587	430	254	111	50	49	158	342	585	761	(9)
(10)		CDD50	2293	2	3	31	87	235	394	525	516	315	156	24	5	(10)
(11)		CDD65	302	0	0	0	1	13	54	110	100	22	2	0	0	(11)
(12)		CDH74	2425	0	0	0	12	99	438	922	809	137	8	0	0	(12)
(13)		CDH80	691	0	0	0	0	9	118	277	264	23	0	0	0	(13)
(14)	Wind	WSAvg	8.2	9.2	9.2	9.0	8.5	8.2	7.9	7.6	7.2	7.3	7.8	8.1	9.0	(14)
(15)	Precipitation	PrecAvg	25.3	2.0	1.9	1.7	1.9	2.4	2.1	2.2	2.2	1.8	2.3	2.2	2.5	(15)
(16)		PrecMax	36.1	5.2	3.9	5.4	5.2	6.6	4.6	6.8	5.0	4.0	5.4	4.4	6.7	(16)
(17)		PrecMin	18.8	0.2	0.2	0.3	0.3	0.1	0.3	0.6	0.3	0.0	0.4	0.5	0.9	(17)
(18)		PrecStd	4.3	1.0	1.0	1.1	1.1	1.3	1.0	1.3	1.1	1.0	1.2	0.9	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	59.2	67.5	75.7	80.8	88.7	92.3	94.0	83.0	74.3	62.1	56.4	(19)
(20)		MCWB	52.7	49.9	52.8	58.4	65.4	69.8	69.5	69.4	65.3	62.7	57.8	53.5		(20)
(21)		2%	DB	53.2	54.0	62.8	70.4	76.8	83.4	86.3	86.9	78.0	69.3	58.9	54.1	(21)
(22)		MCWB	50.7	48.5	51.9	56.0	61.6	66.9	67.4	67.5	63.4	61.0	55.7	51.8		(22)
(23)		5%	DB	51.0	51.8	58.7	65.4	73.0	79.0	82.7	82.3	73.6	65.4	56.3	52.1	(23)
(24)		MCWB	48.6	48.1	50.2	53.8	60.1	64.9	66.1	66.2	62.1	59.1	53.5	50.0		(24)
(25)		10%	DB	48.8	49.9	55.6	61.6	69.1	75.2	79.2	78.3	70.1	62.5	54.2	50.1	(25)
(26)		MCWB	46.6	46.8	49.5	52.1	58.4	63.0	64.9	64.3	60.2	57.7	51.7	48.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	52.0	55.7	60.6	66.8	72.3	71.2	71.7	67.5	64.3	59.1	54.4	(27)
(28)		MCDWB	55.0	55.4	62.1	73.4	78.1	85.2	86.1	86.6	78.0	70.9	61.1	55.6		(28)
(29)		2%	WB	51.1	50.2	53.6	57.1	64.0	68.6	69.4	69.3	65.3	61.8	56.0	52.1	(29)
(30)		MCDWB	52.7	52.8	59.5	66.6	73.7	79.6	82.7	82.4	74.3	67.3	58.1	53.7		(30)
(31)		5%	WB	49.0	48.7	51.9	54.9	61.5	66.3	67.7	67.4	63.4	60.1	53.9	50.3	(31)
(32)		MCDWB	50.7	51.2	57.0	63.2	70.1	75.7	79.4	78.6	71.2	64.6	56.0	52.0		(32)
(33)		10%	WB	46.8	47.0	50.2	53.0	59.3	64.4	66.1	65.6	61.8	58.5	52.0	48.3	(33)
(34)		MCDWB	48.5	49.6	55.0	60.0	67.1	73.5	76.8	75.7	68.5	62.0	54.0	49.9		(34)
(35)	Mean Daily Temperature Range	MDBR	8.0	10.0	12.7	14.8	15.7	16.5	17.1	16.9	15.4	12.2	9.1	8.1		(35)
(36)		5% DB	MCDBR	9.3	12.4	17.6	20.0	20.6	22.1	22.2	22.3	19.7	15.5	10.3	8.9	(36)
(37)		MCWBR	8.1	8.7	9.8	9.7	10.2	10.0	8.6	8.6	8.6	8.8	7.9	7.8		(37)
(38)		5% WB	MCDBR	9.1	11.0	14.3	17.4	18.9	19.2	19.5	19.5	17.2	14.0	9.6	8.8	(38)
(39)		MCWBR	8.4	8.7	9.1	9.4	10.7	10.3	8.6	8.9	8.7	8.7	8.0	8.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.333	0.353	0.387	0.404	0.411	0.413	0.411	0.401	0.375	0.370	0.342	0.326		(40)
(41)		taud	2.388	2.342	2.241	2.216	2.234	2.264	2.262	2.312	2.373	2.372	2.393	2.361		(41)
(42)		Ebn at Noon	228	250	259	267	270	270	268	265	261	240	221	212		(42)
(43)		Edh at Noon	22	28	36	41	42	41	41	37	32	27	21	20		(43)
(44)	All-Sky Solar Radiation	RadAvg	290	538	912	1383	1630	1819	1722	1497	1155	683	351	244		(44)
(45)		RadStd	33	76	113	156	157	170	141	144	96	55	34	37		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page